

## Definite integrals



- 1
- a  $e^2 - e^{-4}$
- b  $\frac{e^{12}}{4} - \frac{1}{4}$
- c  $\frac{2e^3}{3} - \frac{2}{3}$
- d  $\frac{1}{3}e^{12} - \frac{1}{3}e^{-6}$
- e  $-\frac{1}{2}e^{-10} + \frac{1}{2}e^6$
- f  $13 + e^2$
- g  $\frac{2e^{15}}{5} - \frac{e^{-21}}{7} - \frac{9}{35}$
- h  $4e^{\frac{3}{4}} + 15e^{-\frac{3}{5}} - 4e^{-\frac{3}{4}} - 15e^{\frac{3}{4}}$
- i  $\frac{1}{4}e^{16} + 4e^8 - \frac{1}{4}e^{12} - 4e^6 + 16$
- j  $\frac{e^8}{2} + \frac{2e^{12}}{3} + \frac{e^{16}}{4} - \frac{e^{-8}}{2} - \frac{2e^{-12}}{3} - \frac{e^{-16}}{4}$
- k  $-\frac{1}{3}e^{-9} + \frac{1}{2}e^6 + \frac{1}{3}e^{12} - \frac{1}{2}e^{-8}$
- l  $\frac{1}{4}e^{64} - \frac{1}{4}e^4 + 1323$
- 2
- a 1
- b -1
- c  $2\sqrt{3}$
- d -2
- e 0
- f 0
- g  $2\sqrt{3}$
- h  $2\sqrt{3} + 2$
- i  $\frac{6}{\pi}$
- j 0
- k  $\frac{1}{2} - \frac{1}{\sqrt{2}}$
- l 0
- m  $-\frac{\sqrt{3}}{2} + \frac{3}{2}$
- n  $\frac{\sqrt{3}}{4} + 4$
- o -6
- p  $\frac{3}{16}$
- q 0
- 3
- a  $y' = e^x - e^{-x}$
- b  $\ln(e^3 + e^{-3}) - \ln 2$
- 4
- a  $y' = 3xe^{3x}$
- b  $\frac{26e^{27} - 17e^{18}}{9}$

## Graph symmetry

- 5
- a  $\left(\frac{e^8}{4} - \frac{e^{-4}}{4}\right) \text{ units}^2$
- b  $(e^2 - 1) \text{ units}^2$
- c  $(e^5 - 1) \text{ units}^2$
- d  $(e^4 + 15) \text{ units}^2$
- e  $\left(\frac{e^{12}}{6} - \frac{e^{-6}}{6}\right) \text{ units}^2$
- f  $(2e^9 - 2e^{-9}) \text{ units}^2$
- g  $(4e^{\frac{1}{8}} - 4e^{-\frac{1}{8}}) \text{ units}^2$

6

a  $6 \text{ units}^2$

b  $\frac{2}{3} \text{ unit}^2$



c  $1 \text{ unit}^2$

d  $12 \text{ units}^2$

e  $1 \text{ unit}^2$

f  $3 \text{ units}^2$

g  $4 \text{ units}^2$

h  $6\pi \text{ units}^2$

i  $\left(\frac{\pi}{9} + 3\right) \text{ units}^2$

j  $\left(\pi + \frac{1}{4}\right) \text{ units}^2$

k  $(\sqrt{2} + 2) \text{ units}^2$

7

a  $x = 0$

c Odd

e  $\left(\frac{2}{3}e^3 + \frac{2}{3}e^{-3} - \frac{4}{3}\right) \text{ units}^2$

b  $f(x) \rightarrow e^{3x}$

d  $\left(\frac{1}{3}e^9 + \frac{1}{3}e^{-9} - \frac{2}{3}\right) \text{ units}^2$

8

a 0

b i  $0 \leq x \leq \frac{\pi}{12}$  ii  $-\frac{\pi}{12} \leq x \leq 0$

c  $\frac{1}{4} \text{ unit}^2$

9

a  $2\sqrt{2 - \sqrt{2}} \text{ units}^2$

b  $16 \text{ units}^2$

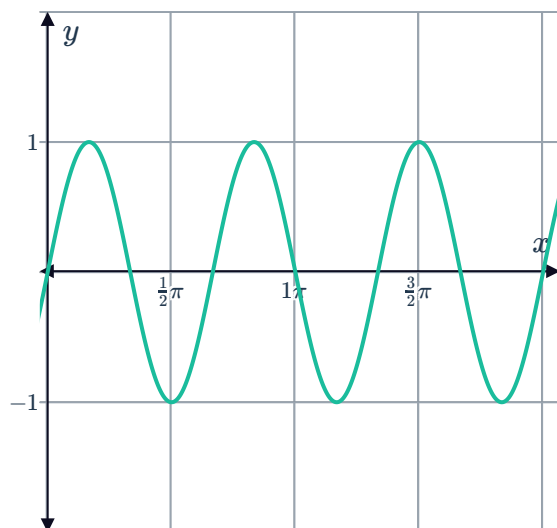
10

a 10 radians

b  $\frac{10\sqrt{2}}{\pi} \text{ units}^2$

11

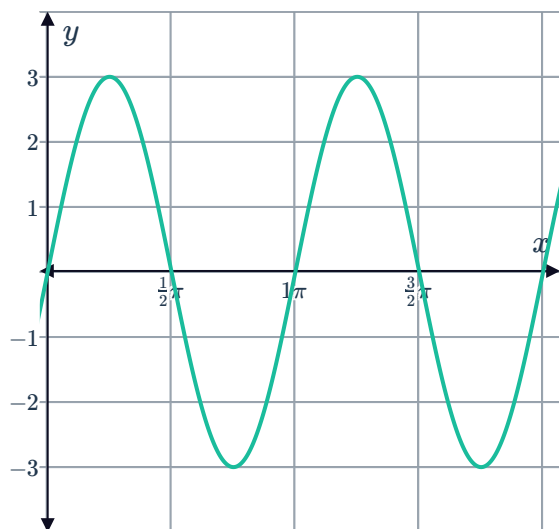
a



b  $\left(\frac{\sqrt{2}}{6} + \frac{1}{3}\right) \text{ units}^2$

12

a



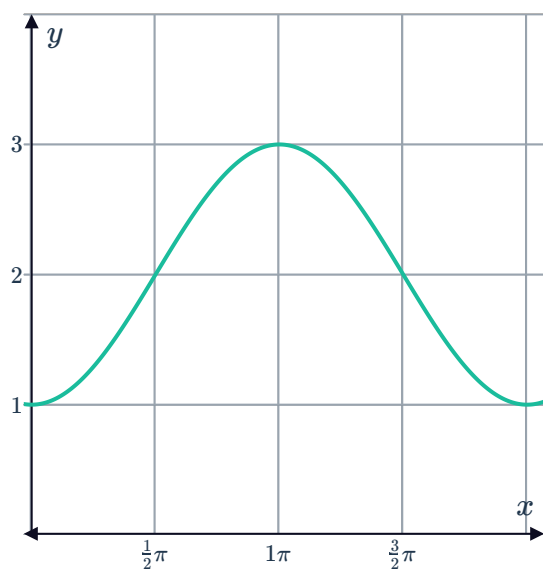
b

i  $3 \text{ units}^2$

ii  $\frac{9}{4} \text{ units}^2$

13

a



b  $(\pi - 1) \text{ units}^2$

Use technology

14 20.61

15  $0.65 \text{ units}^2$